

An Analysis of Protein structure prediction

Protein structure prediction

Protein structure prediction -- Part 1

Secondary structure prediction is a set of techniques in bioinformatics that aim to predict the local secondary structures of proteins based only on knowledge of their amino acid sequence. For proteins, a prediction consists of assigning regions of the amino acid sequence as likely alpha helices, beta strands (often termed extended conformations), or turns. The success of a prediction is determined by comparing it to the results of the DSSP algorithm (or similar e.g. STRIDE) applied to the crystal structure of the protein. Specialized algorithms have been developed for the detection of specific well-defined patterns such as transmembrane helices and coiled coils in proteins. The best modern methods of secondary structure prediction in proteins were claimed to reach 80% accuracy after using machine learning and sequence alignments; this high accuracy allows the use of the predictions as feature improving fold recognition and ab initio protein structure prediction, classification of structural motifs, and refinement of sequence alignments. The accuracy of current protein secondary structure prediction methods is assessed in weekly benchmarks such as LiveBench and EVA.

Protein structure prediction -- Part 2

The α -helix is the most abundant type of secondary structure in proteins. The α -helix has 3.6 amino acids per turn with an H-bond formed between every fourth residue; the average length is 10 amino acids (3 turns) or 10 Å but varies from 5 to 40 (1.5 to 11 turns). The alignment of the H-bonds creates a dipole moment for the helix with a resulting partial positive charge at the amino end of the helix. Because this region has free NH₂ groups, it will interact with negatively charged groups such as phosphates. The most common location of α -helices is at the surface of protein cores, where they provide an interface with the aqueous environment. The inner-facing side of the helix tends to have hydrophobic amino acids and the outer-facing side hydrophilic amino acids. Thus, every third of four amino acids along the chain will tend to be hydrophobic, a pattern that can be quite readily detected. In the leucine zipper motif, a repeating pattern of leucines on the facing sides of two adjacent helices is highly predictive of the motif. A helical-wheel plot can be used to show this repeated pattern. Other α -helices buried in the protein core or in cellular membranes have a higher and more regular distribution of hydrophobic amino acids, and are highly predictive of such structures. Helices exposed on the surface have a lower proportion of hydrophobic amino acids. Amino acid content can be predictive of an α -helical region. Regions richer in alanine (A), glutamic acid (E), leucine (L), and methionine (M) and poorer in proline (P), glycine (G), tyrosine (Y), and serine (S) tend to form an α -helix. Proline destabilizes or breaks

an α -helix but can be present in longer helices, forming a bend.

Protein structure prediction -- Part 3

The peptide bonds in the chain are polar, i.e. they have separated positive and negative charges (partial charges) in the carbonyl group, which can act as hydrogen bond acceptor and in the NH group, which can act as hydrogen bond donor. These groups can therefore interact in the protein structure. Proteins consist mostly of 20 different types of L- α -amino acids (the proteinogenic amino acids). These can be classified according to the chemistry of the side chain, which also plays an important structural role. Glycine takes on a special position, as it has the smallest side chain, only one hydrogen atom, and therefore can increase the local flexibility in the protein structure. Cysteine in contrast can react with another cysteine residue to form one cystine and thereby form a cross link stabilizing the whole structure. Protein structure arises from a sequence of secondary structure elements, such as α helices and β sheets. In secondary structures, regular patterns of H-bonds are formed between the main chain NH and CO groups of spatially neighboring amino acids, and the amino acids have similar Φ and Ψ angles.

Protein structure prediction -- Part 4

β -sheets are formed by H-bonds between an average of 5–10 consecutive amino acids in one portion of the chain with another 5–10 farther down the chain. The interacting regions may be adjacent, with a short loop in between, or far apart, with other structures in between. Every chain may run in the same direction to form a parallel sheet, or in the reverse direction to form an antiparallel sheet, or the chains may form a mixed sheet. The pattern of hydrogen bonding is different in the parallel and antiparallel configurations. Each amino acid in the interior strands of the sheet forms two H-bonds with neighboring amino acids, whereas each amino acid on the outside strands forms only one bond with an interior strand. Looking across the sheet at right angles to the strands, more distant strands are rotated slightly counterclockwise to form a left-handed twist. The C α -atoms alternate above and below the sheet in a pleated structure, and the R side groups of the amino acids alternate above and below the pleats. The Φ and Ψ angles of the amino acids in sheets vary considerably in one region of the Ramachandran plot. It is more difficult to predict the location of β -sheets than of α -helices. The situation improves somewhat when the amino acid variation in multiple sequence alignments is taken into account.

Protein structure prediction -- Part 5

Historical perspective To date, over 20 different secondary structure prediction methods have been developed. One of the

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first algorithms was Chou–Fasman method, which relies predominantly on probability parameters determined from relative frequencies of each amino acid's appearance in each type of secondary structure. The original Chou-Fasman parameters, determined from the small sample of structures solved in the mid-1970s, produce poor results compared to modern methods, though the parameterization has been updated since it was first published. The Chou-Fasman method is roughly 50–60% accurate in predicting secondary structures. The next notable program was the GOR method is an information theory-based method. It uses the more powerful probabilistic technique of Bayesian inference. The GOR method takes into account not only the probability of each amino acid having a particular secondary structure, but also the conditional probability of the amino acid assuming each structure given the contributions of its neighbors (it does not assume that the neighbors have that same structure). The approach is both more sensitive and more accurate than that of Chou and Fasman because amino acid structural propensities are only strong for a small number of amino acids such as proline and glycine. Weak contributions from each of many neighbors can add up to strong effects overall. The original GOR method was roughly 65% accurate and is dramatically more successful in predicting alpha helices than beta sheets, which it frequently mispredicted as loops or disorganized regions. Another big step forward was using machine learning methods. First artificial neural networks methods were used. As a training sets they use solved structures to identify common sequence motifs associated with particular arrangements of secondary structures. These methods are over 70% accurate in their predictions, although beta strands are still often underpredicted due to the lack of three-dimensional structural information that would allow assessment of hydrogen bonding patterns that can promote formation of the extended conformation required for the presence of a complete beta sheet. PSIPRED and JPRED are some of the most known programs based on neural networks for protein secondary structure prediction. Next, support vector machines have proven particularly useful for predicting the locations of turns, which are difficult to identify with statistical methods. Extensions of machine learning techniques attempt to predict more fine-grained local properties of proteins, such as backbone dihedral angles in unassigned regions. Both SVMs and neural networks have been applied to this problem. More recently, real-value torsion angles can be accurately predicted by SPINE-X and successfully employed for ab initio structure prediction.